

 ALBUKHARY INTERNATIONAL UNIVERSITY	SCHOOL OF COMPUTING AND INFORMATICS CCE2333- Software Testing
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**Module 1: Understanding and Testing Control Flow
Constructs in C Programming
(Report)**

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Marks:

Part A	Part B	Part C	Pard D

Part A: Defined Test Scenario and Test Case

Program Name	Test Scenario Description	Test Case ID	Input(s)	Expected Output
if_else_test.c	odd number	TC_IF_ELSE_1	3	The number 3 is odd
	even number	TC_IF_ELSE_2	4	The number 4 is even
	zero	TC_IF_ELSE_3	0	the number 0 is even
	negative even number	TC_IF_ELSE_4	-7	the number -7 is odd
switch_test.c	Add	TC_SWITCH_1	23, 2	sum=25
	MULTIPLY	TC_SWITCH_2	20, 10	Difference =10
	SUBSTRACT	TC_SWITCH_3	5, 5	product =25
	DIVIDE	TC_SWITCH_4	0,5	invalid choice
while_loop_test.c	odd number	TC_WHILE_1	8	8 is an even number 9 is an odd number 10 is an even number 11 is an odd number 12 is an even number
	even number	TC_WHILE_2	2	2 is an even number 3 is an odd number 4 is an even number

				5 is an odd number 6 is an even number
	zero	TC WHILE 3	0	no output
	negative	TC WHILE 4	-3	no output

do_while_test.c	even number	TC_DO_WHILE_1	2	2 is a even number 3 is a odd number 4 is a even number 5 is a odd number 6 is a even number
	zero	TC_DO_WHILE_2	0	0 is a even number 1 is a odd number 2 is a even number 3 is a odd number 4 is a even number
	negative number	TC_DO_WHILE_3	-7	-7 is addd number
	odd number	TC_DO_WHILE_4	3	3 is a odd number 4 is a even number 5 is a odd number 6 is a even number 7 is a odd number
for_loop_test.c	Loop from 1 to 5	TC_FOR_1	-	2 is a even number 3 is a odd number 4 is a even number 5 is a odd number
	-	TC_FOR_2	-	
	-	TC_FOR_3	-	
	-	TC_FOR_4	-	

Part B: Test Run

ID	Expected Output	Actual Output	Pass/Fail
TC_IF_ELSE_1	The number 3 is odd	The number 3 is odd	pass
TC_IF_ELSE_2	The number 4 is even	The number 4 is even	pass
TC_IF_ELSE_3	the number 0 is even	the number 0 is even	pass
TC_IF_ELSE_4	the number -7 is odd	the number -7 is odd	pass
TC_SWITCH_1	sum=25	sum=25	pass
TC_SWITCH_2	Difference =10	Difference =10	pass
TC_SWITCH_3	product =25	product =25	pass
TC_SWITCH_4	invalid choice	invalid choice	pass
TC_WHILE_1	8 is an even number 9 is an odd number 10 is an even number 11 is an odd number 12 is an even number	8 is an even number 9 is an odd number 10 is an even number 11 is an odd number 12 is an even number	pass
TC_WHILE_2	2 is an even number 3 is an odd number 4 is an even number 5 is an odd number 6 is an even number	2 is an even number 3 is an odd number 4 is an even number 5 is an odd number 6 is an even number	pass
TC_WHILE_3	no output	no output	pass
TC_WHILE_4	no output	no output	pass
TC_DO_WHILE_1	2 is a even number 3 is a odd number 4 is a even number 5 is a odd number 6 is a even number	2 is a even number 3 is a odd number 4 is a even number 5 is a odd number 6 is a even number	pass
TC_DO_WHILE_2	0 is a even number 1 is a odd number 2 is a even number 3 is a odd number 4 is a even number	0 is a even number 1 is a odd number 2 is a even number 3 is a odd number 4 is a even number	pass
TC_DO_WHILE_3	-7 is a odd number	-7 is aodd number	pass
TC_DO_WHILE_4	3 is a odd number 4 is a even number 5 is a odd number 6 is a even number 7 is a odd number	3 is a odd number 4 is a even number 5 is a odd number 6 is a even number 7 is a odd number	pass
TC_FOR_1	Loop from 1 to 5	2 is a even number 3 is a odd number 4 is a even number 5 is a odd number	pass

TC FOR 2			
TC FOR 3			
TC FOR 4			pass
			pass
			pass

Part C: Reflection Questions

Q1: What did you observe when using the `do...while` loop with a negative number?

 its giving only one output and also its right out output as the logic is met with the requirements and code is right actually

Q2: What happens when you input a non-integer (e.g., character) into the program?

 when we put non integer its giving an unpredictable result as we are using %d . scan f.

Q3: How does the switch structure handle invalid choices or division by zero?

 Invalid choices (any menu number that doesn't match a `case`) are handled by the `default` case, which prints an error message like "`Invalid choice`"

Q4: What is the key difference between `while` and `do...while` loops in terms of condition checking?

 in while loop the condition start at first but in do while the condition start at the last

Part D: Reflection Questions

Summarize your understanding of control flow constructs and basic testing:

It was good learning as I have seen how the code works and the logic followed in the code / and I have tried different input to see if any errors in the logic of code but the code is ok , it works fine with the logic . In the future I may apply this knowledge for software testing .

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-End of Module 1-